

PALLETPILOT · REQUIREMENTS

PalletPilot requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	R15 prototype mass limit restated as 45 kg and R7 tied to ISO 3691-4, decided by Amish on 2026-10-02 (PLP-DEC-001)
0.7	2026-10-01	Amish Chadha	TRL 4 verification note for the steering range decided by Amish (PLP-DDR-003 A1); no requirement changed
0.6	2026-10-01	Amish Chadha	Constructable design (PLP-DDR-003): mass, release and cost figures; R17 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Record decisions PLP-DDR-001 (R7 redefined for the lidar layer, R17 to \$1,550, 1,500 kg on level floors); status from PLP-CAL-001
0.2	2026-09-25	Amish Chadha	

First measurable requirements for TRL 2, with status against the concept estimates

0.1	2026-09-24	Amish Chadha	Initial scaffold
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PalletPilot requirements

These requirements are checked by calculation in PLP-CAL-001. Eleven of eighteen requirement lines are met on paper, one is not met (R5 bearing accuracy), two are at risk (R7, R9), three cannot be verified at TRL 3 (R1, R6, R16), and R17 is reported against its value-engineering target. Targets are still to be validated with users after site visits (see PLP-PRB-001). Decisions recorded in PLP-DDR-001 on 2026-09-25 changed R2, R4, R7 and R17; decisions in PLP-DDR-002 on the same date restated R4 and R8 (bumper-only and untested follow mode at 0.15 m/s) and R9 (two contactors in series). On 2026-09-26 Amish approved the budget of \$1,610 (PLP-DDR-002); on 2026-10-01 he set budgets as value-engineering targets, not limits, so R17 is reported as over or under the target. On 2026-10-01 the design was made constructable (PLP-DDR-003, Draft): the kit mass and cost rose and the release lever was resized; no requirement changed status. On 2026-10-02 Amish restated R15's prototype mass limit as 45 kg, which the kit meets, and decided that follow mode is a driverless truck function under ISO 3691-4 (R7) (PLP-DEC-001).

The **design load case** is a 1,000 kg pallet on a 64 kg donor jack with a kit of about 44.4 kg, about 1,108 kg in total, on a level, dry, sealed concrete floor.

The **design duty** is one 8 h shift of 60 pallet moves, each 40 m loaded and 40 m empty, with four starts in each direction.

Table 1. Requirements.

ID	Requirement	Target	Verification	Status (PLP-CAL-001)
R1	Fit a common manual jack	Bolts to 27 x 48 in (685 x 1220 mm), 2,500 kg manual jacks with a common steering yoke; no welding, cutting or drilling of load-bearing parts; fitted in 2 h or less and removed in 1 h or less with hand tools	Survey of three donor models; fitting sequence review	Not verifiable at TRL 3; the model assumes one donor geometry
R2	Move the design load	Start and move 1,000 kg on a level floor; kit rating 1,500 kg at 0.8 m/s or less on level floors, below the donor rating (decided, PLP-DDR-001 D7)	Traction and torque calculation	Met: 25.5 N·m per motor needed of 30 N·m specified; 750 N traction. A 1,500 kg ramp start is outside the rating
R3	Handle short ramps	Start, climb and hold the design load on a 2 % grade	Traction, torque and brake calculation	Met: start needs 488 N of 750 N (margin 1.54, 1.23 at friction 0.4)
R4	Walking-pace speeds	Walk mode 1.2 m/s (4.3 km/h) or less handle-end first and 0.8 m/s or less forks first; creep 0.3 m/s or less with the handle near upright; follow mode 0.6 m/s or less, and 0.15 m/s or less until the lidar layer is built and tested (decided, PLP-DDR-001 D6; interim limit lowered from 0.2 m/s to the bumper-only limit by PLP-DDR-002)	Controller limits; later timed runs	Met by design (software limits)
R5	Follow the operator	In follow mode, hold a 1.5 m gap within ± 0.3 m and a bearing within $\pm 10^\circ$ (2σ) of the tag at walking pace	UWB error budget; later tracking trials	Not met: gap ± 0.06 m, but bearing about $\pm 16^\circ$ with a 450 mm anchor baseline and filtering
R6	Stop if the operator is lost	Stop command within 0.3 s of losing the tag link, a gap over 3 m, a tag stop press or the tag leaving a $\pm 45^\circ$ cone	Firmware design review; later bench test	Not verifiable at TRL 3; timing budget 0.20 s
R7	Stop before hitting a person in follow mode	The lidar layer detects a person in the travel path and the truck stops without contact from 0.6 m/s with the design load, with the bumper as the last layer (redefined for option 1A, D1)	Stopping calculation; later test with an ISO 3691-4 test piece; follow mode is treated as a driverless truck function under ISO 3691-4 (decided 2026-10-02), so personnel detection and stopping must meet that standard	At risk: stop 0.42 m level and 0.76 m worst case, inside a 0.86 m protective field; the sensor is not safety-rated and detection of dark clothing is unproven
R8	Stop on bumper contact	Bumper contact removes drive power and applies the brakes within 100 ms; any operation that relies on the bumper alone limited to 0.15 m/s or less (decided, PLP-DDR-002; was 0.2 m/s)	Stop chain design review	Met on paper: 0.10 s chain; 0.15 m/s stops in 38 mm of the 40 mm edge travel
R9	Emergency stop	Two hardwired emergency stops (tiller head and enclosure) plus the tag stop;	Stop chain review; PL calculation	At risk: dual-channel relay and two contactors

ID	Requirement	Target	Verification	Status (PLP-CAL-001)
		the hardwired stops cut drive power through a dual-channel safety relay and two main contactors in series (decided, PLP-DDR-002), independent of software; target performance level PL d under ISO 13849-1		in series; PL not yet calculated
R10	Park safely	Brakes apply with power off and hold the design load on a 2 % grade	Brake torque check against datasheet	Met on paper: 40 N·m specified against 21.7 N·m needed; datasheet to confirm
R11	Keep the operator in control	Walk mode drives only with the handle between about 20° and 70° from vertical; belly-reverse paddle on the tiller head reverses the truck when pressed	Design review; model check	Met: the handle clears the enclosure by 40 mm at 70° and 14 mm when lowered flat
R12	Work a full shift	Complete the design duty on one charge with 20 % or more of usable energy left	Energy calculation	Met: 293 Wh used of 410 Wh usable (28 % left)
R13	Charge overnight	Full charge in 5 h or less from a 120 V or 230 V outlet through a certified charger	Charger datasheet	Met: 4.5 h at 5 A
R14	Push by hand when unpowered	One lever lifts the drive wheels clear in 10 s or less (brakes need no release once the wheels are clear); unpowered push force no more than 10 % above the bare jack	Design review; later push-force test	Met on paper: +4.2 % push force; peak lever effort about 58 N; drive wheels lift about 10 mm (PLP-DDR-003)
R15	Keep the jack usable	Kit mass 40 kg or less as the goal, 45 kg or less for the prototype (decided 2026-10-02, PLP-DEC-001); overall length increase 300 mm or less at floor level; lowered fork height and pallet entry unchanged	Mass estimate; model check	Met for the prototype: 290 mm added length (met); 44.4 kg, 0.6 kg under the 45 kg prototype limit and 4.4 kg over the 40 kg goal; rechecked at the motor quote
R16	Indoor environment	Operate at 0 to 40 °C; electronics and connectors IP54; no charging below 0 °C	Datasheets and design review	Not verifiable at TRL 3; set by the specification of bought parts
R17	Affordable	Kit parts cost against the value-engineering target of \$1,610, donor jack excluded (budget decided, D2; \$1,610 approved by Amish, 2026-09-26, PLP-DDR-002; a target, not a limit, 2026-10-01)	Priced BOM (bom/bom.csv)	Value-engineering target: USD 1,610. Estimated cost of the constructable design: USD 1,690 (USD 80 over the target)

■ Requirements not met or at risk

- **R5, follow bearing accuracy, is not met.** Filtering brings the error to about $\pm 16^\circ$; $\pm 10^\circ$ would need a 706 mm anchor baseline, wider than the jack. Amish decided to evaluate angle-of-arrival modules and lidar leg tracking at TRL 4, which is on hold (PLP-DDR-002).
- **R15 mass is met for the prototype** (45 kg limit) and misses the 40 kg goal by 4.4 kg. The 30 N·m class hub motors weigh about 7 kg each; the second contactor adds 0.3 kg; the parts added to make the design constructable (clamp jaw, drive arms, pivot pin, cam shaft and lever) add 2.4 kg (PLP-DDR-003). Amish accepted 2.0 kg over for now, to be rechecked at the motor quote (PLP-DDR-002); on 2026-10-02 he accepted 44.4 kg for the prototype and set R15's prototype limit at 45 kg, with about 1.5 kg of savings (5 mm top plate, tube lever, hollow pivot pin) carried into the TRL 4 drawings (PLP-DEC-001). The 40 kg goal is still not met.
- **R17, cost, is USD 80 over its value-engineering target:** USD 1,690 against USD 1,610. Savings worth trying are listed in PLP-DEC-001.
- **Steering range (no requirement covers it; verified at TRL 4).** With the kit fitted the jack steers about 40° each way, set by rubber stops, against about 90° for a bare jack, for a turning radius of roughly 1.5 m about the load wheels (PLP-DDR-003, P9). Amish accepted this range for the prototype on 2026-10-01 (PLP-DDR-003, A1). Verification: the TRL 4 aisle test, a go or no-go gate. Pass when the loaded kit makes a right-angle turn from an aisle into a standard pallet bay; if it cannot, the drive layout changes before TRL 5 (PLP-DEC-001).
- **R7 is at risk.** The stopping distances fit the lidar field on paper, but the sensor is not safety-rated, so this cannot support an ISO 3691-4 claim. On 2026-10-02 Amish decided that follow mode is treated as a driverless truck function under ISO 3691-4, so personnel detection and stopping functions must meet that standard; it is reclassified as an operator-controlled assist mode only if a standards body or a notified body confirms that the walking follower counts as the operator (PLP-DEC-001).
- **R9 is at risk** until the stop chain, now with two contactors in series, has a PL calculation.

■ Assumptions

- Rolling resistance 1.2 % and breakaway resistance 2.5 % for polyurethane steer and load wheels on smooth concrete. These are assumptions to be measured on a real jack.
- Tread-to-floor friction coefficient 0.5 for the drive wheels on dusty concrete, 0.4 as a low case; lower on wet or polished floors.
- The drive wheels carry 1.5 kN of spring preload and no share of the pallet load.
- Motor plus driver efficiency 70 %; controls, UWB, lidar and lights draw about 9.2 W in total.
- Stop delays of 0.25 s for the lidar layer and 0.10 s for the bumper and emergency stops.
- R8 is met by the 0.15 m/s speed limit, not by a longer-travel edge; a firmware or hardware fault that lets the truck exceed 0.15 m/s in a bumper-only mode would defeat it.
- The legal status of the retrofit is set by OSHA 1910.178(a)(4) (D8). Follow mode is treated as a driverless truck function under ISO 3691-4 (decided 2026-10-02), the stricter answer that R7 and R9 anticipated. The kit is documented for research use in a closed area (D8).